Statistical Frameworks

Today's agenda:

Discuss statistical approaches,

R code basics

Discuss the modelling process,

Start working with code

Statistical Approaches

How many ways are there to do statistics?

How many ways are there to do statistics?

Three

Frequentist
Likelihood
Bayesian

This course will focus mostly on likelihood

One dataset, three answers

Bolker provides a dataset from Duncan and Duncan (2000)

Seeds at 941 stations, Kibale, Uganda
Two species: *Polyscias* (tiny seeds) and *Pseudospondias* (~50 g seeds)
Seeds taken at 51 stations.

Does being taken depend on species?

Frequentist

Probability = long-run frequency

Assume nothing is going on. How surprising is what we saw?

Answer: a p-value

Likelihood

Probability = probability of the data, given the parameters

Which parameter values make what we saw most probable?

Answer: an estimate, with support

Bayesian

Probability = degree of belief

Start with a prior, update it with the data

Answer: a posterior distribution

Same data, same conclusion

Frequentist (Fisher's exact test) $p = 3.8 \times 10^{-6}$

Likelihood (MLE + likelihood ratio) estimate 3.62, limits 2.13 – 6.16

Bayesian (flat prior) mode 3.48, credible interval 2.01 – 6.01

None of the intervals include 1

The frameworks differ in what the numbers *mean*

Different Statistical Paradigms

Aspect	Frequentist Statistics	Likelihood	Bayesian Statistics
Interpretation of probability	Long-run frequency of events	Probability of data, given parameters	Degree of belief / uncertainty
Parameters	Fixed but unknown	Fixed, but unknown	Random variables with distributions
Data	Random (due to sampling)	Fixed; the thing whose probability we evaluate	Fixed, updates beliefs about parameters
Prior information	Not used	Not used	Required (can be informative or weak/noninformative)
Main output	Point estimates, confidence intervals, p-values	Point estimates, likelihood-based confidence limits	Posterior distributions, credible intervals, Bayes factors
Confidence vs. credibility	95% CI: in repeated samples, 95% of intervals contain the true value	Support interval: parameter values that make the data nearly as probable as the best estimate	95% credible interval: given data + prior, 95% chance parameter lies in the interval
Inference process	Hypothesis tests, maximum likelihood, resampling	Maximise the likelihood, compare models by likelihood ratio test	Bayes' theorem: Prior × Likelihood → Posterior
Computation	Often analytic formulas, asymptotics, bootstraps	Numerical optimisation	Often simulation (e.g., MCMC, variational inference)
Common critique	Ignores prior knowledge, interpretation of p-values can be misleading	No probability statement about the parameter itself; relies on asymptotics in small samples	Can be subjective (depends on choice of prior), computationally intensive
Example question	"If we repeated this study many times, how often would we see results like this?"	"Which parameter values make my data most probable?"	"Given the data, what is the probability the hypothesis is true?"

Questions so far?

(Note: this stuff will become more clear as we go)

R Code Basics

Let's actually use it

Open RStudio

Make a new R script

Save it in your project (the one created last time)

Note: Type along. Don't paste (typing helps you memorize things better)

R as a calculator

In R, type:

```
2 * 8
```

Then run it (run button, enter key, CTRL + SHIFT +R or CMD + SHIFT + R)

What does the output look like?

R as a calculator

In R, type:

```
sqrt(25)
```

Then run it

What does the output look like?

What does [1] refer to?

Assignment

Type and run:

```
x <- sqrt(36)
x
```

What does this do?

Why is nothing shown when you type `x <- sqrt(36)` ?

(Note the book uses = for assignment. `<-` (read: gets) is now standard practice)

Getting help

R has multiple ways to get help.

Try these:

```
?sqrt

example(sqrt)

help.search("correlation")
```

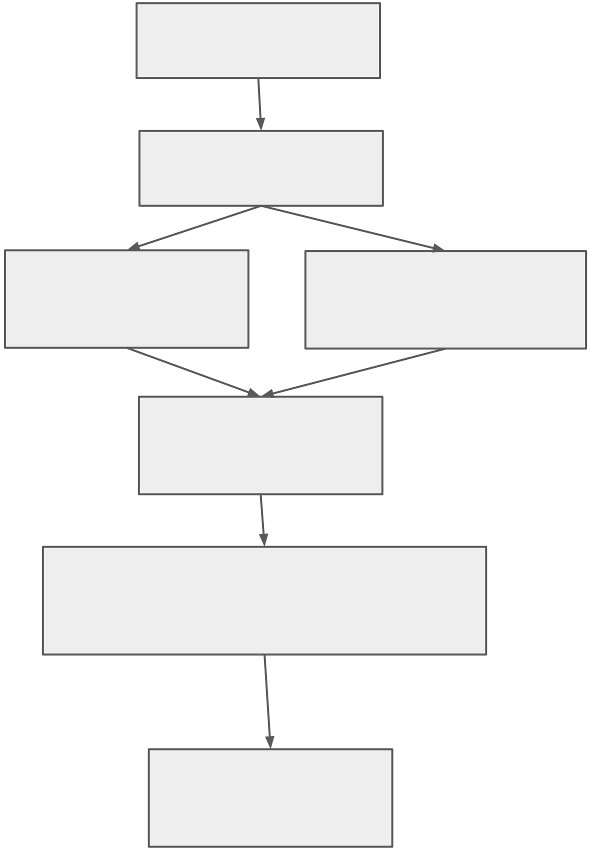
What’s different about them? What’s similar?

Getting help

We’ll go into this more as we go, but these are a good place to start

The modelling process

What is the modelling process?



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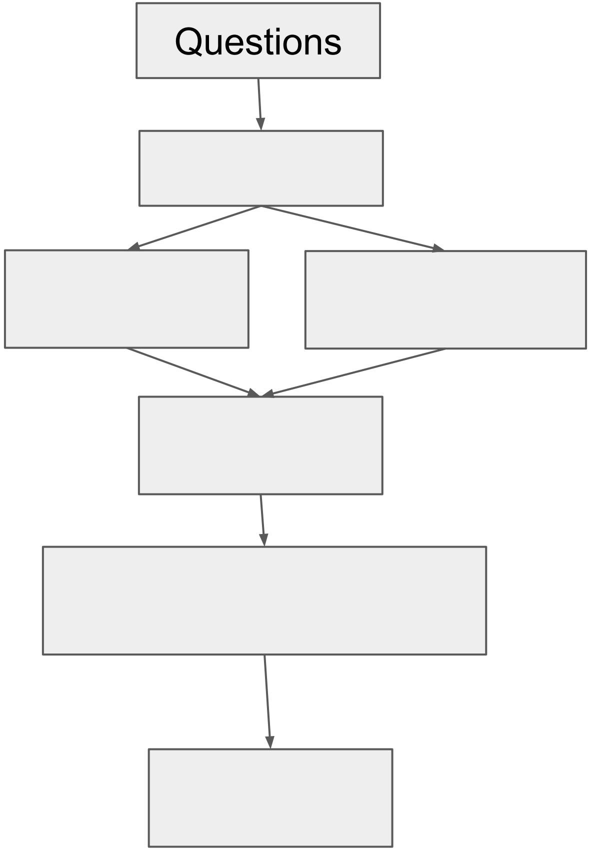

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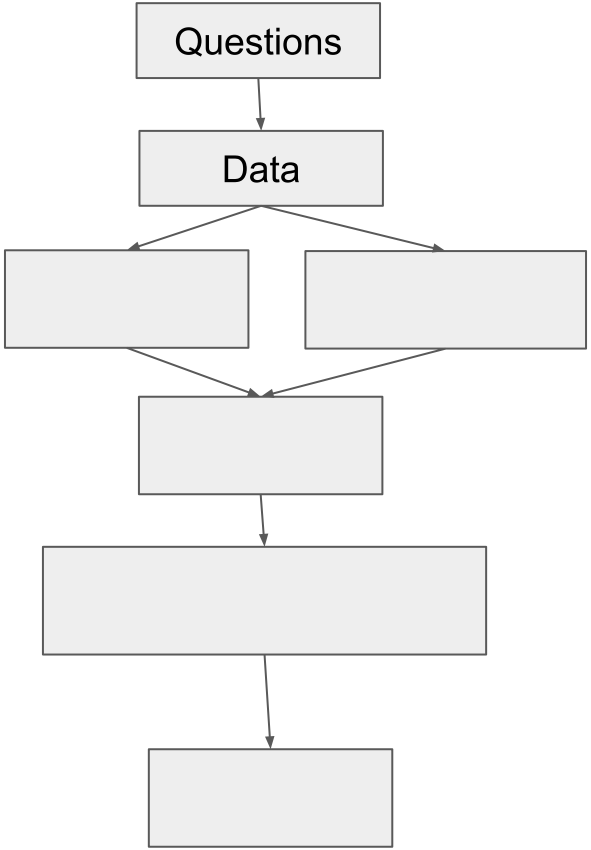

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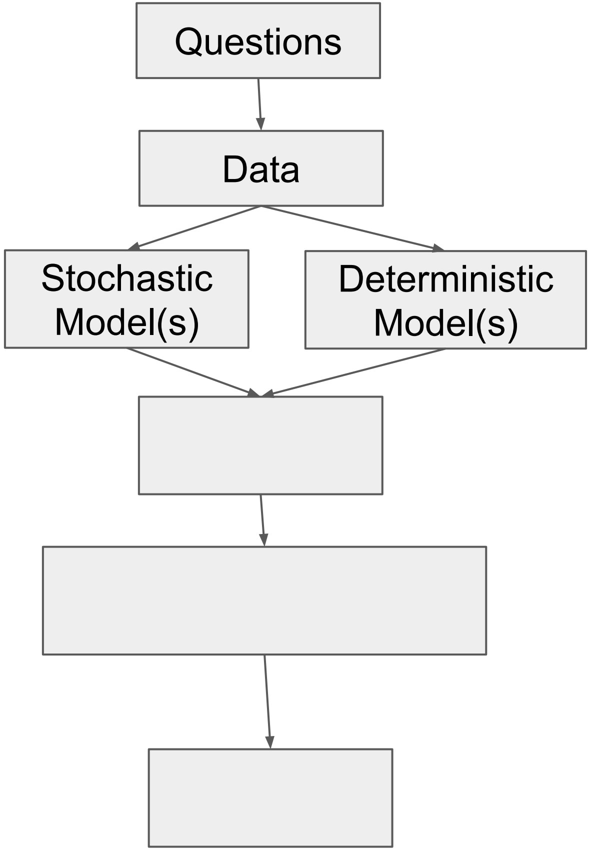

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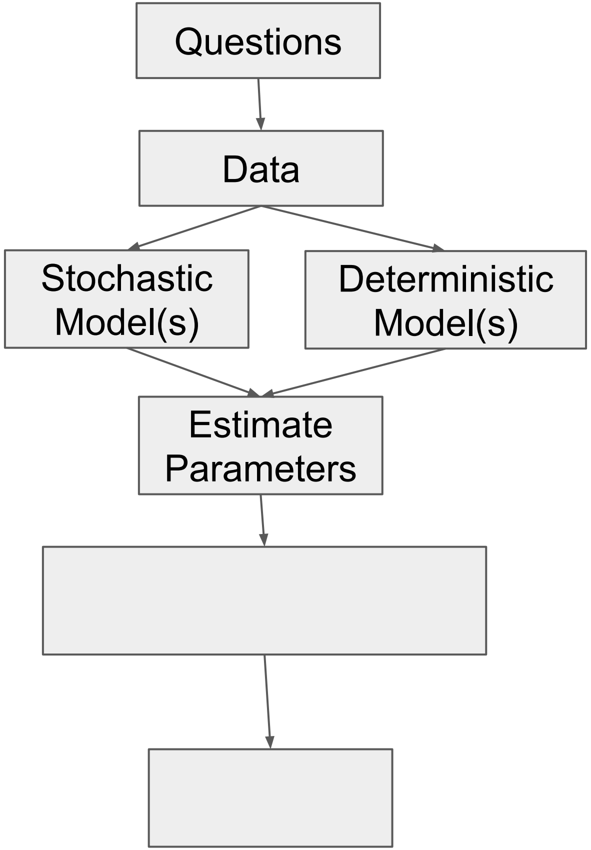

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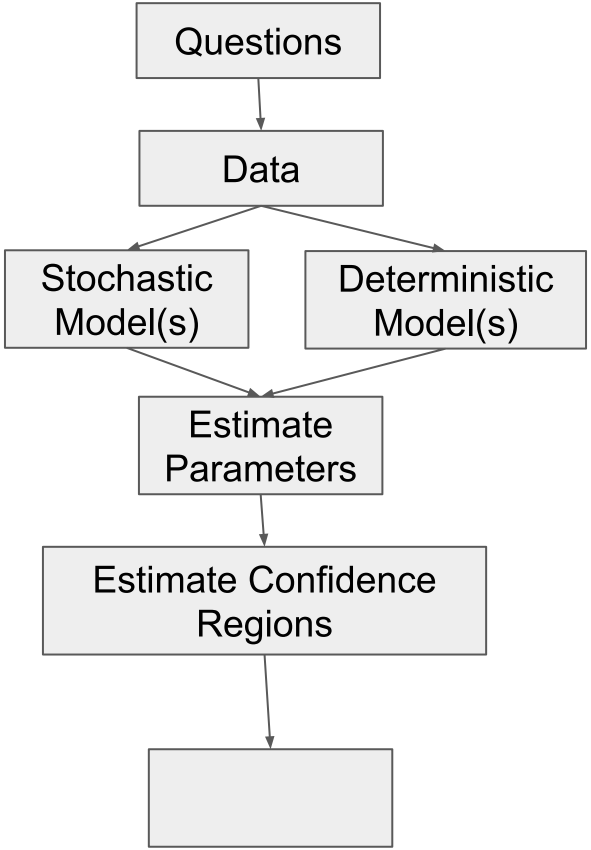

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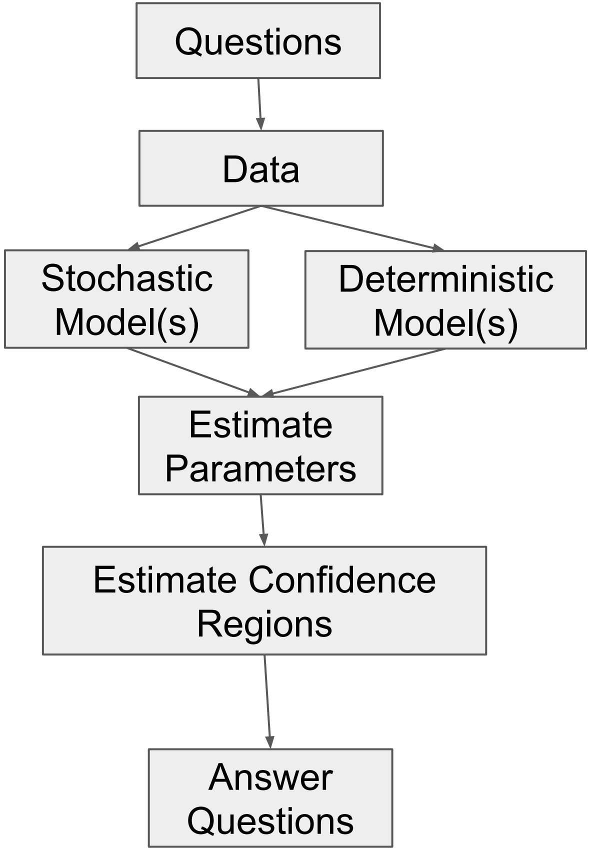

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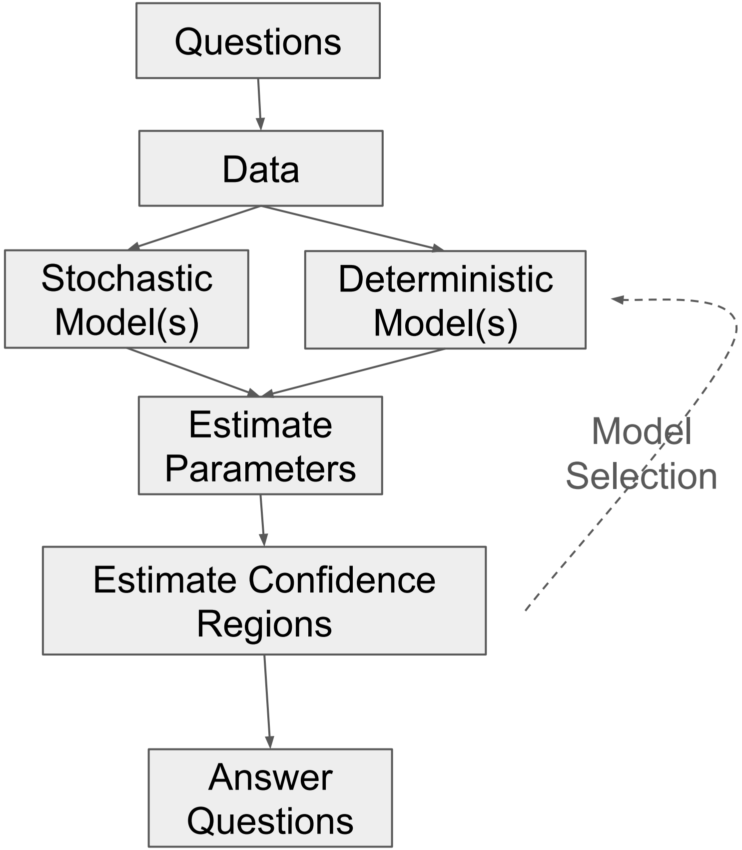

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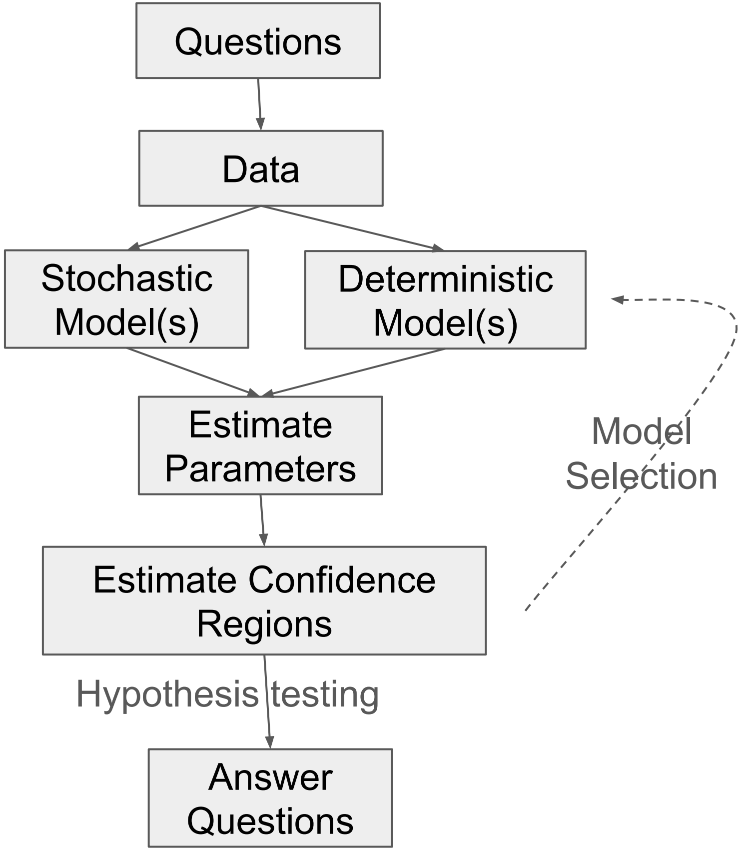

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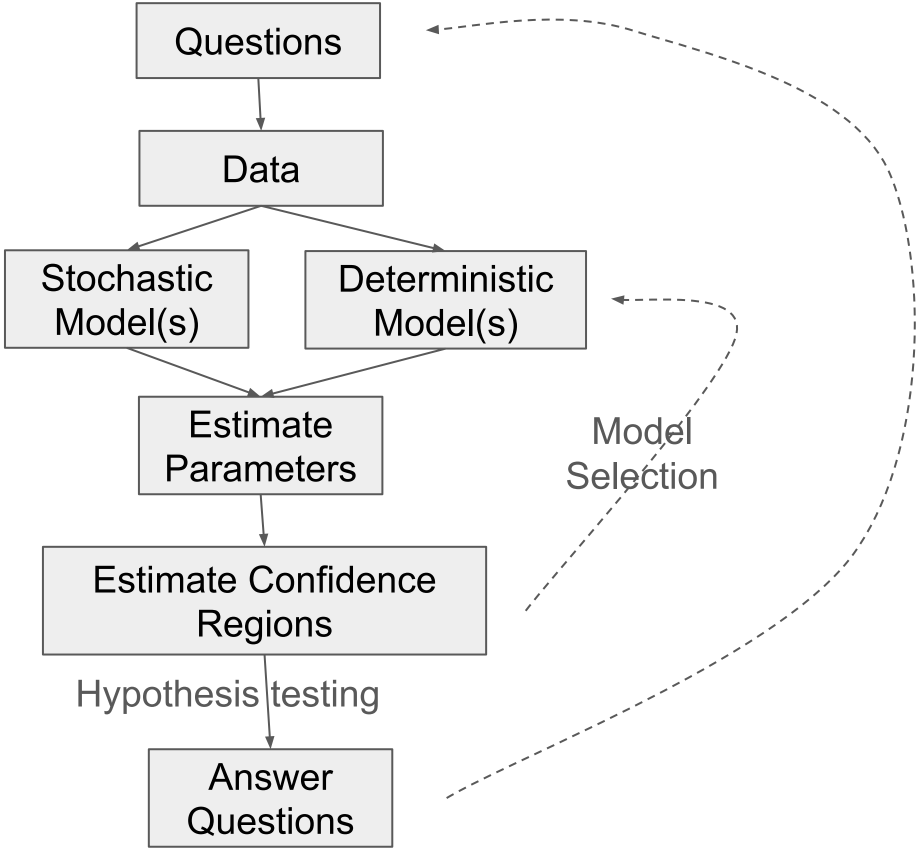


Asking a good question

Need to consider two levels:

General: does disease select against cannibalism in tiger salamanders?

Specific: what is the difference in probability of becoming a cannibal between populations A and B?

Asking a good question

Need to consider two levels:

General: does disease select against cannibalism in tiger salamanders?

Specific: what is the difference in probability of becoming a cannibal between populations A and B?

General is more aligned with how you TALK about the question

Specific is more aligned with how you TEST the questions

Too vague, too specific

Too vague: "I want to explore the population genetics of cannibalism"

Too specific: "What is the difference in the intercepts of these two linear regressions?"

Too vague can make things hard to test

Too specific can lead to uninteresting questions

Your turn

Take the question you thought about last week.

Write it at the other level:

general if you had specific, specific if you had general.

Swap with a neighbour.

Can they tell which is which?

Questions so far?

(Again, this will become more clear as we go along)

Code: build, fit, test

First we'll play nature

We’ll build a dataset where we know the true answer

mean = 2 * frogs: the real signal

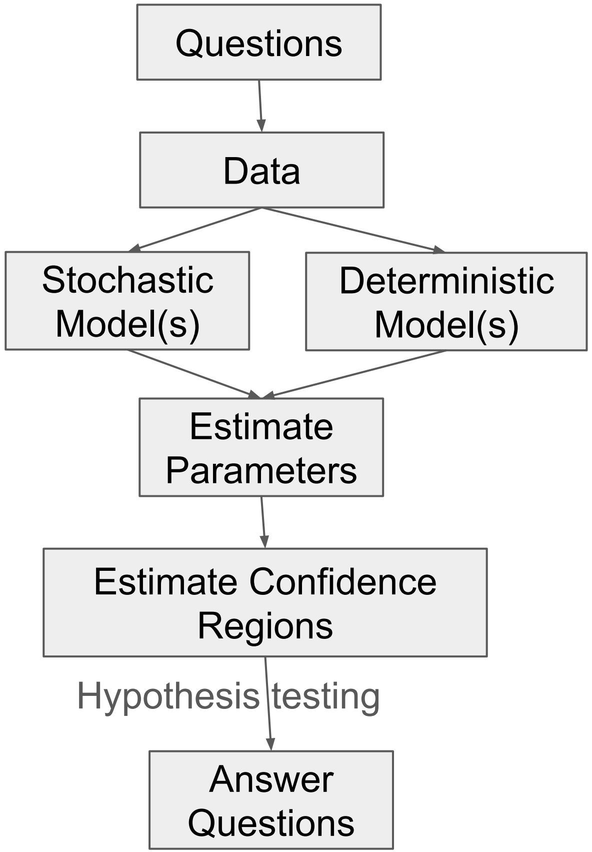
sd = 0.5: the real noise

Then we'll forget we know, and analyse it

Let's walk the flowchart

We'll build a dataset where we know the true answer

Then see whether our analysis can recover it



The signal

```
frogs <- c(1.1, 1.3, 1.7, 1.8, 1.9, 2.1,
           2.3, 2.4, 2.5, 2.8, 3.1, 3.3,
           3.6, 3.7, 3.9, 4.1, 4.5, 4.8,
           5.1, 5.3)
```

“frogs” is a **vector** of length 20 describing adult frog density in 20 populations

The noise

```
set.seed(101)

tadpoles <- rnorm(n = 20,

                  mean = 2 * frogs,

                  sd = 0.5)
```

mean = 2 * frogs is the **deterministic** model

sd = 0.5 is the **stochastic** model

Look at it

```
plot(x = frogs, y = tadpoles)

abline(a = 0, b = 2)
```

The line is the truth. Why don't the points sit on it?

Turn the noise up

```
tadpoles_noisy <- rnorm(20, 2 * frogs, sd = 3)

plot(frogs, tadpoles_noisy)

abline(a = 0, b = 2)
```

Same relationship (2 * frogs). Would you have spotted it this time?

Can we recover the truth?

```
coef(lm(tadpoles ~ frogs))

coef(lm(tadpoles_noisy ~ frogs))

confint(lm(tadpoles ~ frogs))
```

The real slope is 2. Which one got closer?

Note: you just fit two linear models!

One draw proves little

Let’s replicate that simulation 1000 times:

```
slopes_clean <- replicate(1000,
  coef(lm(rnorm(20, 2*frogs, sd = 0.5) ~ frogs))[2])

slopes_noisy <- replicate(1000,
  coef(lm(rnorm(20, 2*frogs, sd = 3) ~ frogs))[2])
```

One draw proves little

Now let’s summarize what we found:

```
mean(slopes_clean)

mean(slopes_noisy)

sd(slopes_clean)

sd(slopes_noisy)
```

What did adding noise do to our findings?

Our interval missed

`confint(frog_lm)` → 1.69 to 1.98

The true slope is 2. Not in there.

Build 1000 intervals this way and 95% of them contain the truth.

Ours was one of the other 5%.

Summarize and test

```
mean(tadpoles)

summary(tadpoles)

cor(frogs, tadpoles)

cor.test(frogs, tadpoles)
```

Is that p-value impressive?

Run it again

```
set.seed(202)

tadpoles2 <- rnorm(20, 2 * frogs, sd = 0.5)

cor(frogs, tadpoles2)
```

How did this differ? Why?

Different data. Same truth.

This is demographic stochasticity, from last week.

What we just did

Data: 20 populations

Deterministic model: the line lm() fits

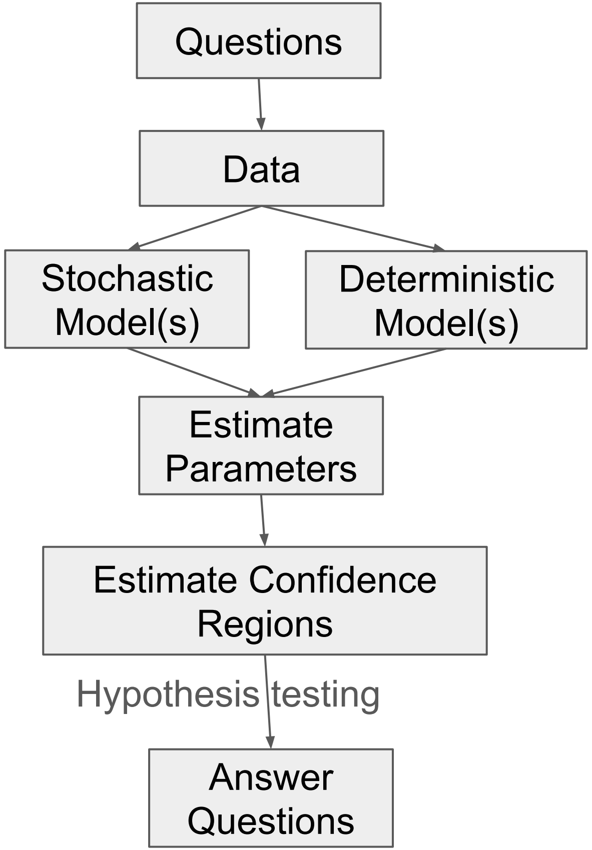
Stochastic model: the normal errors it assumes

Estimate parameters: coef()

Confidence regions: confint()

Hypothesis testing:cor.test()

lm() never saw how we made the data



Next time:

Before class:

- read 2.1 - 2.3
- Think about types of data you might want to look at

During class:

- Discuss 2.1 - 2.3
- Work on loading in data